

Project Title

SummarIQ – AI-Powered PDF Summarizer Application for Scientific Papers

Project Overview

Objective

The primary goal of this project is to develop a simplified AI system that summarizes research papers in PDF format. The system will extract main points, including technical content like equations, tables, and key findings. For the first semester, the focus will be on text extraction and generating concise summaries using pretrained NLP models.

Scope

In Scope (Semester 1):

- Upload a PDF research paper and extract text using Python libraries.
- Identify and retain key technical content, including equations.
- Generate extractive or abstractive summaries using pretrained NLP models.
- Display the summarized text in a clean interface.
- iOS demo app: Simple interface to upload PDF, process, and show summary .

Out of Scope (Semester 1):

- Handling images, graphs, or charts in summaries.
- Deep understanding of equations (semantic evaluation).
- Multi-document summarization or literature survey generation.
- Real-time summarization or cloud deployment.

Limitations:

- Summaries may omit fine details from complex equations.
- Works best on text-heavy PDFs.
- Dependent on quality of PDF formatting.

AI Techniques and Tools

Core AI Methods (Semester 1):

- **Text Extraction:** GROBID, *PyPDF2*, *pdfplumber* for PDF → text conversion.
- **Preprocessing:** Remove headers, footers, page numbers; normalize text.
- **Summarization Models:**
 - Pretrained Transformers: *T5-small*, *BART* for abstractive summaries.
 - Extractive summarization: *LexRank* or *sumy* for selecting key sentences.

Programming Libraries:

- `transformers` → NLP models
- `PyPDF2`, `pdfplumber` → PDF text extraction
- `NumPy`, `Pandas` → data processing
- `SwiftUI` → iOS interface
- `Flask` → optional backend API connecting Python model to iOS app

Semester Milestone:

- Extract text from research paper PDFs.
- Identify sections and equations (as LaTeX strings or placeholders).
- Generate concise summaries (3–5 sentences).
- Display summary in an iOS app interface (upload → processing → result).
- Stretch goal: Highlight equations or key points in the summary.

Future Scope:

- **Equation Awareness:** Integrate tools to parse and semantically understand equations.
- **Multi-Document Summarization:** Summarize multiple papers into one literature review.
- **Graph & Table Summarization:** Include tables, charts, and images in summaries.
- **Deep Learning Enhancements:** Fine-tune transformer models on research paper datasets.
- **iOS Deployment:** Convert model to Core ML for offline summarization on iPhone.

Data Plan:

- **Custom Data:** Select 10–20 PDFs from arXiv or other open-access research papers for testing/demo.
- **Public Datasets:**
 - **ScisummNet** – NLP summaries of scientific papers.
 - **SurveySum** – Summaries of multiple scientific papers.
 - **MS2** – Large-scale dataset with documents and summaries.
 - **cPAPERS** – Papers containing equations and tables for summarization.

Stakeholders:

Project Team:

- Satyabrata Das – AI Developer & iOS Developer → Implement AI pipeline, app interface, and integration.

End Users:

- Researchers & students → Quickly read research papers.

- Educators → Summarize content for lectures.

Other Stakeholders:

- Universities → Can use lightweight summarization tools.
- Open-access repositories → Provide research PDFs.

Self-Rating of Confidence (Beginner Stage):

- Transformers (Hugging Face): 3/10 → New to NLP fine-tuning.
- PDF Processing (PyPDF2, pdfplumber): 7/10 → Comfortable extracting text.
- iOS/SwiftUI: 6/10 → Able to create simple app interfaces and integrate backend.

Computing Infrastructure:

Objective and Scope

The primary goal of this infrastructure is to support a pipeline for extracting text from research paper PDFs, preserving mathematical expressions (as LaTeX or [Equation] tokens), and generating concise 3–5 sentence summaries using abstractive or extractive methods. The infrastructure must enable fast, cost-effective, and reproducible experimentation and deployment across development, training, and demo phases.

System Objectives and Functional Tasks

Core Tasks

1. **Text Extraction and Cleanup** – Extract textual content from PDFs, removing headers, footers, and pagination using `pdfplumber` or `PyPDF2`.
2. **Equation Detection and Preservation** – Identify inline and block-level math content; store them as LaTeX strings or [Equation] placeholders.
3. **Summarization** –
 - **Abstractive**: Use `T5-small` or `BART-base` fine-tuned on scientific text.
 - **Extractive fallback**: Apply `LexRank` when abstractive models underperform.
4. **iOS Demo Application** – Enable file upload, cloud-based text processing, and display of summaries with equation highlights.

Data Types

- **Primary**: Plain text from PDFs, including LaTeX math tokens.
- **Secondary**: Metadata (title, author list, abstract) and text-based tables.
- **Out of Scope (Semester 1)**: Figures, semantic equation parsing, and image extraction.

Quantitative Performance Benchmarks

Metric	Target	Justification / Notes
Latency (Inference)	≤ 2 sec (GPU), ≤ 10 sec (CPU)	Ensures interactivity in demo settings using T5-small or BART-base.
Throughput (Server)	5–20 requests/minute per GPU	Supports small concurrent user base; scalable horizontally.
Summarization Quality	ROUGE-1: 25–40	Realistic for fine-tuned small transformer models on limited data.
Accuracy Validation	$\geq 80\%$ human evaluator pass rate	Ensures summaries capture main paper content.
Cost (Training + Inference)	$\leq \$50$ /month (demo phase)	Achievable using spot GPU instances (A10/T4).

Performance Validation

- Evaluate summary quality using **ROUGE-1/2/L** metrics.
- Measure **p50/p95 latency** using API logs during test runs.
- Conduct **load testing** with tools like *Locust* or *JMeter* to simulate concurrent requests and measure auto-scaling thresholds.

Hardware and Resource Planning:

Component	Minimum Spec (Development)	Recommended Spec (Training)	Justification
CPU	8 vCPUs	16 vCPUs	For parallel data processing during training.
RAM	32 GB	64 GB	Required for PDF parsing and model fine-tuning.
GPU (VRAM)	12–16 GB (T5-small/BART-base)	40+ GB (BART-large)	Ensures smooth fine-tuning and inference.
Storage	200 GB SSD	1 TB SSD (if caching datasets)	For datasets, model checkpoints, and logs.

Example GPU Options

- **Local:** NVIDIA RTX 3090 / 4090
- **Cloud:** NVIDIA A10, T4, A100 instances on AWS/GCP

Decision:

A single GPU node (T4/A10 class) will serve both training (spot/preemptible instances) and inference (on-demand, single-GPU endpoint). Larger models (e.g., BART-large) may use A100-class hardware if budget allows.

Software Environment:

Layer	Tools / Frameworks	Purpose
OS	macOS / Ubuntu	Standard ML development environment
Frameworks	PyTorch + Hugging Face Transformers	Core for T5/BART fine-tuning
PDF Processing	pdfplumber, PyPDF2	Text and math extraction
NLP Preprocessing	regex, NLTK, spaCy	Tokenization, sentence segmentation
Summarization Models	T5-small, BART-base, LexRank	Abstractive and extractive summarization
Backend	Flask / FastAPI + gunicorn + uvicorn	API inference server
Containerization	Docker	Reproducibility and deployment consistency
Frontend	SwiftUI (iOS)	User-facing upload and summary display

Deployment Plan:

Phase	Platform	Description
Development	Local GPU / VM	Training small models and pipeline validation.
Semester 1 (Local Host)	Local Server / Flask	Initial deployment and testing of the Fast API locally before cloud deployment.
Demo Deployment	Cloud GPU instance (AWS/GCP)	Fast API with GPU-backed inference endpoint.
Storage	S3 / GCS	PDF uploads and processed summaries.
Database	PostgreSQL / Firebase	User metadata and summary caching.
Future Expansion	Core ML conversion	Enable on-device inference in later semesters.

Tradeoff Analysis:

Factor	Option A (T5-small)	Option B (BART-base)	Tradeoff Summary
Model Size	60M params	140M params	BART-base yields

			higher quality but higher cost/latency.
Inference Latency (GPU)	~1.5 sec	~3.5 sec	T5-small preferred for interactive demo.
GPU Memory Requirement	~12 GB	~24 GB	BART-base may require a higher-tier GPU.
Monthly Cloud Cost	~\$25	~\$60	Cost scales with GPU class and runtime.
ROUGE-1 Score (Expected)	30–35	35–45	Slightly higher accuracy at higher resource cost.

Decision:

For Semester 1, T5-small is selected for its balance between cost, latency, and quality. BART-base will be considered for later improvements if resource constraints allow.

Scalability and Load Validation:

Scaling Strategy

- Horizontal scaling: Multiple Flask/FastAPI replicas behind a load balancer.
- Auto-scaling policy: Trigger new instances when GPU utilization >70% or latency >3 sec (p95).
- Batching and caching: Use cached summaries for repeated PDFs; batch small requests to improve GPU utilization.

Load Testing

- Use Locust or JMeter to simulate concurrent users and validate scalability.
- Measure performance under 5, 10, and 20 concurrent requests to confirm that average latency remains under 2 seconds for GPU-backed inference.

Risk Management and Triggers:

Risk	Trigger Condition	Mitigation Strategy
Latency Degradation	Median latency > 2 sec or p95 > 5 sec	Apply quantization (8-bit), enable mixed precision, or upgrade GPU instances.
Budget Overrun	Monthly cloud spend > 25% of plan	Switch to preemptible instances or reduce model complexity.
Low ROUGE / Quality	ROUGE-1 < 25 or <70% human pass rate	Fine-tune longer or apply domain-specific data augmentation.
Scalability Limit	>20 concurrent users degrade performance	Enable horizontal auto-scaling and response caching.

Security, Privacy, and Ethics (Trustworthiness):

Trustworthiness is embedded throughout the AI lifecycle, ensuring that the summarization model remains **fair, transparent, privacy-preserving, and secure**. Each stage of the system's lifecycle — from problem definition to post-deployment — explicitly considers ethical and technical safeguards.

Problem Definition

Strategy: Identify ethical and fairness risks before system design begins.

Plan: Examine possible harms such as biased summaries, factual distortion, or the omission of critical details. Assess how these risks may affect users like students, researchers, or publishers.

Tool/Method: Apply the **Data Ethics Canvas** to document potential risks, stakeholder concerns, and mitigation actions.

Measurable Criteria: Completion of an ethical-risk checklist prior to data collection.

Data Collection

Strategy: Use representative, unbiased, and privacy-conscious data sources.

Plan: Collect papers from **arXiv** and similar open repositories, remove sensitive metadata (e.g., author names), and balance datasets across domains.

Technical Implementation:

- **Snorkel:** Generate synthetic data to reduce underrepresentation of specific paper types.
- **IBM Diffprivlib:** Apply differential privacy to protect metadata.
Metrics: Monitor dataset balance using **domain representation ratios** and **KL-divergence**.

Model Development

Strategy: Build fair, explainable, and robust summarization models.

Plan: Fine-tune **BART** models while assessing fairness across scientific domains.

Technical Implementation:

- **Fairlearn:** Compare summarization quality (ROUGE, F1) across disciplines such as physics, medicine, and computer science.
- **SHAP:** Explain model attention and detect systematic bias in how sentences are weighted.
Audit Criteria: Retraining is triggered if the fairness gap (e.g., ROUGE difference across domains) exceeds **10%**.

Deployment

Strategy: Deploy the summarization system securely and transparently.

Plan: Package the model using **Docker** and serve via **FastAPI** with **JWT-based authentication** and HTTPS. Log API activity for traceability.

Tool: BentoML for serving, versioning, and monitoring.

User Transparency:

- Provide a **Model Card** describing data sources, limitations, and update frequency.
- Display a short disclaimer such as: “*Summaries may omit technical nuances — please verify with the original paper.*”

Monitoring and Maintenance

Strategy: Continuously evaluate fairness, drift, and performance post-deployment.

Plan:

- Use **NannyML** to detect data or concept drift monthly.
- Run quarterly fairness audits with **Fairlearn**.
- Gather user feedback through rating surveys to guide retraining.

Metrics Tracked:

- **Quality:** ROUGE-L, factual consistency (FactCC).
- **Fairness:** Inter-domain performance differences, sentiment balance.
- **Drift:** Population Stability Index (PSI).
Retraining Policy: Retrain when $PSI > 0.05$ or quality metrics fall by more than 5%.
Transparency: Publish changelogs and performance updates after each model revision.

Human–Computer Interaction (HCI)

For this project, HCI principles are applied throughout the design process to ensure the **LaTeX summarization system** is simple, accessible, and valuable for students, researchers, and educators. User needs and feedback drive every stage — from requirement gathering to post-deployment evaluation.

Understand User Requirements

Approach:

- Conducted short Google Form surveys with students and faculty to understand how they currently read and summarize LaTeX-based research papers.
- Followed up with semi-structured interviews (3–5 students, 1–2 faculty members) to explore pain points and expectations.

Insights Expected:

- Users want faster ways to skim LaTeX papers and extract main ideas.
- Equations and mathematical notations must be preserved accurately in summaries.
- Users prefer a lightweight, mobile-friendly interface for easy access.

Tool:

Used **Miro Affinity Diagramming** to group feedback into categories such as “*time-saving*,” “*clarity of equations*,” and “*ease of reading*.”

Create Personas and Scenarios

Personas

1. **Student Persona – “Riya”**

Profile: 20-year-old undergraduate computer science student.

Motivation: Needs short, clear summaries for assignments and LaTeX-based project papers.

Pain Points: Reading long .tex documents with equations is tedious on small screens; I want quick, structured insights.

Context of Use: Uses the mobile app between classes or during study sessions.

2. **Researcher Persona – “Arjun”**

Profile: 26-year-old PhD student in Artificial Intelligence.

Motivation: Needs precise summaries that retain equations, LaTeX syntax, and domain-specific content.

Pain Points: Most existing summarizers fail to handle LaTeX markup or skip equations entirely.

Context of Use: Uses a desktop/laptop while analyzing technical LaTeX manuscripts.

3. **Educator Persona – “Dr. Meera”**

Profile: 40-year-old computer science lecturer.

Motivation: Prepares lecture materials from LaTeX-based research papers.

Pain Points: Reviewing multiple .tex files before class preparation is time-consuming.

Context of Use: Uses a web or desktop interface during lecture preparation or research review sessions.

Scenarios

- **Scenario 1:** Riya uploads a .tex file, and the system generates a 3–5 sentence summary readable in under one minute.
- **Scenario 2:** Arjun tests whether mathematical equations are accurately preserved and formatted in the summary.
- **Scenario 3:** Dr. Meera uses generated summaries to quickly identify key research papers for her lectures.

Conduct Task Analysis

Main Task: Upload a LaTeX file and receive a concise, accurate summary with preserved equations.

Subtasks:

1. Open app → Tap “**Upload LaTeX File.**”

2. Select the desired .tex file from device storage.
3. Wait during “**Processing**” with progress feedback displayed.
4. View the generated summary containing text and equations.
5. *(Optional)* Copy, share, or save the summary for later use.

Method:

Used **Hierarchical Task Analysis (HTA)** to break down actions, identify potential friction points (e.g., file handling or loading delays), and streamline task flow.

Identify Accessibility Requirements

Goal:

Design for inclusivity and ensure the summarization system is usable by individuals with diverse abilities and language backgrounds.

Accessibility Strategies:

- Large, high-contrast buttons and readable fonts for mobile and desktop.
- Screen reader support for visually impaired users (e.g., VoiceOver, TalkBack).
- Keyboard navigation and focus indicators for web version accessibility.
- Clear labeling of equations and math expressions for assistive technologies.
- **Language support:** Future implementation for multilingual summaries (English, Hindi, Spanish).

Evaluation Tools:

Use **WAVE** and **Axe** to verify **WCAG 2.1 compliance** in future web-based versions.

Outline Usability Goals (Semester 1)

Building on initial project targets, the following **measurable usability goals** are defined for Semester 1:

Metric	Target	Measurement Method
Task Completion Time	Users should generate and read LaTeX summaries within 1 minute	Observed during usability testing
Task Success Rate	At least 90% of users complete “upload → summary read” without error	Log user interactions
User Satisfaction	Minimum 80% satisfaction in post-test survey	5-point Likert-scale feedback form
Equation Retention Accuracy	At least 85% of equations preserved correctly in output	Manual comparison with source LaTeX
Time Efficiency	Reduce time to understand a LaTeX paper by 50% compared to manual reading	Pre/post comparison study

Evaluation Plan

- Conduct usability testing with **5–7 participants** (mix of students and researchers experienced with LaTeX).
- Measure **task completion time**, **task success rate**, **equation accuracy**, and **user satisfaction scores**.
- Collect both **quantitative metrics** and **qualitative comments**.
- Use feedback to refine:
 - Summary formatting for equations
 - Interface layout and clarity
 - Performance and feedback timing

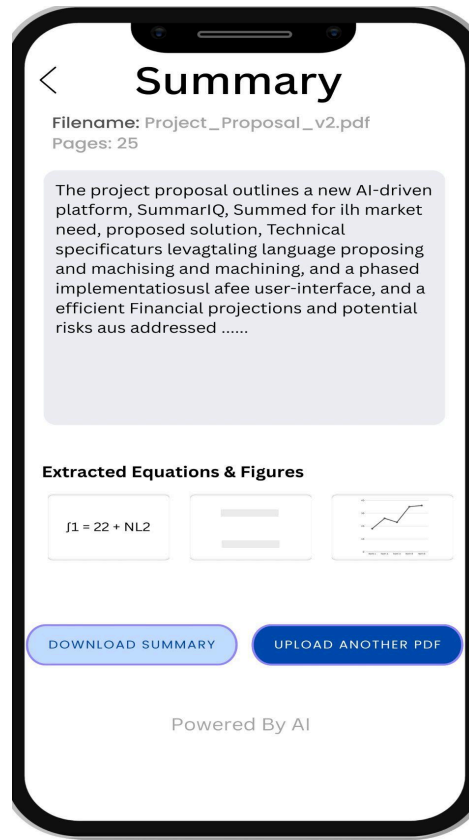
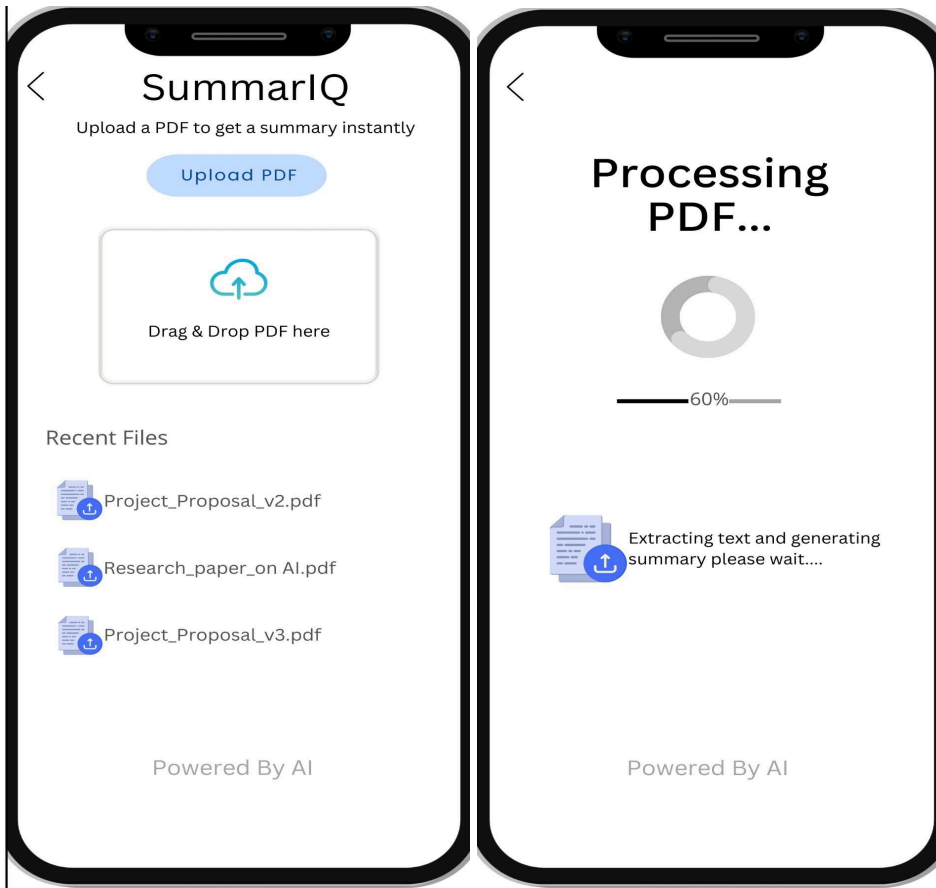
Iteration and Continuous Improvement

- Analyze survey and test data using **descriptive statistics** and **thematic grouping**.
- Revise interface components after each test cycle based on user insights.
- Plan follow-up usability testing at the end of each semester to measure improvement trends.
- Document all changes and feedback to maintain transparency and support **future audits** or publication of user study findings.

Git Repo link:

<https://github.com/Satyabratadas/SummarIQ.git>

User Interface(IOS):



Data Collection Management and Report :

Data Type

Type of Data:

The project primarily manages **unstructured text data** extracted from research papers in **LaTeX format**. Each paper includes textual content, mathematical equations, and structural metadata such as sections (e.g., Introduction, Methods, Results, Conclusion).

The raw data (LaTeX files) are converted into **semi-structured JSON** files containing key-value mappings of sections and content, then loaded into **pandas DataFrames** for further processing.

Data Granularity:

- **Raw Data:** Original LaTeX files downloaded from arXiv.
- **Processed Data:** JSON files containing cleaned, structured text and equations.

Challenges:

- Handling embedded LaTeX math syntax and citations.
- Parsing errors due to non-standard document structures.

Adjustments:

Custom Python parsing scripts were written to detect section headers and LaTeX environments (e.g., `\section{}`, `\begin{equation}`) to ensure structured conversion.

Data Collection Methods

Source of Data:

- **Primary Source:** [arXiv.org](https://arxiv.org) — a reliable and publicly available repository of research papers.
- The dataset includes scientific documents from AI, NLP, and computer vision categories.
- The reliability of arXiv is high due to its established academic standards.

Methodologies Applied:

- **Automated collection using Python scripts** via the arxiv API and web requests.
- Downloaded .tex source files were stored locally for preprocessing.
- **Parsing scripts** converted the data into structured JSON using regex-based section extraction.

Ingestion for Training:

- Data was read from JSON into **pandas DataFrames**.
- Custom **DataLoader** classes and **ETL pipelines** were implemented for preprocessing, cleaning, and section-wise tokenization.

- The data currently resides in local storage in both JSON and DataFrame formats.

Ingestion for Deployment (Planned):

- During deployment, SummarIQ will ingest PDF or LaTeX input via a **REST API endpoint**.
- Preprocessing will be handled in real time on the server using batch processing queues.
- Data will be stored in a cloud database (e.g., AWS S3 or MongoDB) for future retraining.

Compliance with Legal Frameworks

Applicable Laws and Standards:

- **GDPR (General Data Protection Regulation):** Although arXiv data is publicly available, compliance with privacy protection and consent principles is maintained.
- **CC BY License:** arXiv papers used comply with the **Creative Commons Attribution license**, allowing reuse for research with citation.
- **ISO/IEC 27001:** Data is stored securely and locally to maintain confidentiality and integrity.

Compliance Strategy and Results:

- All data was collected only from publicly available open-access papers.
- No personal data or private repositories were accessed.
- License metadata (authors, titles) was retained in the JSON files for attribution.
- Python scripts maintain logs for download verification and integrity checking.

Data Ownership and Access Rights

Ownership and Access Control:

- **Data Ownership:** arXiv retains original paper rights under open-access licensing; SummarIQ uses them under fair research usage.
- **Access Control:** Only authorized project developers can access the local and cloud data stores.
- **Security Measures:**
 - Password-protected local directories.
 - Versioned backups.
 - Future plan: API authentication tokens and two-factor verification for deployment access.

Lessons Learned:

- Implementing directory-level access restrictions reduced accidental overwriting.
- Data access logs improved traceability of collection operations.

Metadata Management

Metadata Content and Management System:

Each processed LaTeX file includes essential metadata fields extracted during parsing:

- **Paper Title** – extracted from the LaTeX header (`\title{}`)
- **Author Name(s)** – extracted from the LaTeX author field (`\author{}`)
- **File ID (UUID)** – a unique identifier automatically generated on the device for each parsed file.

This minimal metadata structure ensures that each document can be uniquely tracked, referenced, and managed throughout preprocessing and model training.

Management Approach:

- Metadata is embedded within the **JSON file** alongside the parsed paper content.
- Each entry includes the UUID as the key field to ensure consistent linking between content and metadata.
- A **pandas DataFrame** stores all metadata records for quick querying and verification during analysis.
- Additionally, a consolidated **metadata index file (CSV)** maintains a list of all processed paper UUIDs to avoid duplication and enable fast lookup.

Challenges and Solutions:

- **Challenge:** Some LaTeX files lacked clear or well-formatted metadata fields (e.g., missing `\author` or `\title` tags).
- **Solution:** Implemented fallback logic in the parser to assign "Not Available" for missing fields while ensuring that a **UUID** is always generated. This guarantees that every file maintains a consistent metadata record even when partial information is missing.

Data Versioning

Version Control System and Strategy:

- **Version Control:** Git + DVC (Data Version Control) used for tracking changes in scripts and datasets.
- **Strategy:**
 - Each preprocessing stage (raw → parsed → cleaned) is saved as a separate dataset version.
 - Git commits tagged with dataset version numbers (e.g., `v1.0_raw`, `v1.1_cleaned`).
 - Metadata updates tracked through commit messages for transparency.

Data Preprocessing, Augmentation, and Synthesis

Preprocessing Techniques:

- **Text Cleaning:** Removal of LaTeX commands, citations, and formatting symbols.

- **Tokenization:** Sentence and word-level tokenization using NLTK and spaCy.
- **Normalization:** Lowercasing and stopwords removal.
- **Equation Extraction:** Extracted math expressions separately to retain structural integrity.
- **Section Segmentation:** Extracted by regex patterns (`\section{}`, `\subsection{}`) for context-based summarization.

Challenges & Solutions:

- Parsing failed for papers with nested sectioning — handled via recursive regex functions.
- Ensured minimal data loss using manual inspection of a subset.

Data Augmentation and Synthesis:

- **Back-translation** for sentence variation during model training.
- **Synonym replacement** to increase linguistic diversity.
- Over-augmentation avoided to maintain the scientific tone of papers.

Data Management Risks and Mitigation

Identified Risk	Mitigation Strategy	Effectiveness
Data corruption during download	SHA-256 hash verification for each file	High
Duplicate or missing papers	Index-based verification and cross-check	High
Parsing failure	Logging errors and reprocessing flagged files	Moderate
Unauthorized access	Local password protection and access logs	High
License violation	Verification of arXiv license before inclusion	High

Data Management Trustworthiness and Mitigation

Trustworthiness Strategies:

- **Data Provenance Tracking:** Maintained download URLs and timestamps for traceability.
- **Reproducibility:** All scripts versioned with consistent preprocessing parameters.
- **Integrity Checks:** Used checksums for every downloaded file.
- **Transparency:** Documented data pipeline in README and Git logs.

Effectiveness and Improvements:

- Provenance tracking ensures reproducibility for future researchers.
- Plan to integrate automated audit logs and digital signatures for enhanced trust.

